

Geometry
Unit 7
Lesson H1 Practice

Name Answer Key
Date _____
Period _____

In $\triangle B K X$, $m\angle KBX = 19^\circ$, $m\angle B X K = 41^\circ$, and $XK = 32$. Determine the measurement of each part of $\triangle B K X$. Show your work below the table. If necessary, round to the nearest tenth.

	Part	Measurement
1.	$m\angle XKB$	120°
2.	$\overline{XB}$	85.1
3.	$\overline{KB}$	64.5

$$\begin{aligned}
 1) \quad & m\angle XKB = 180 - (19 + 41) = 120^\circ \\
 2) \quad & \frac{\sin 19}{32} = \frac{\sin 120}{XB}; \quad XB = 32 \frac{\sin 120}{\sin 19} = 85.121 \\
 3) \quad & \frac{\sin 19}{32} = \frac{\sin 41}{KB}; \quad KB = 32 \frac{\sin 41}{\sin 19} = 64.484
 \end{aligned}$$

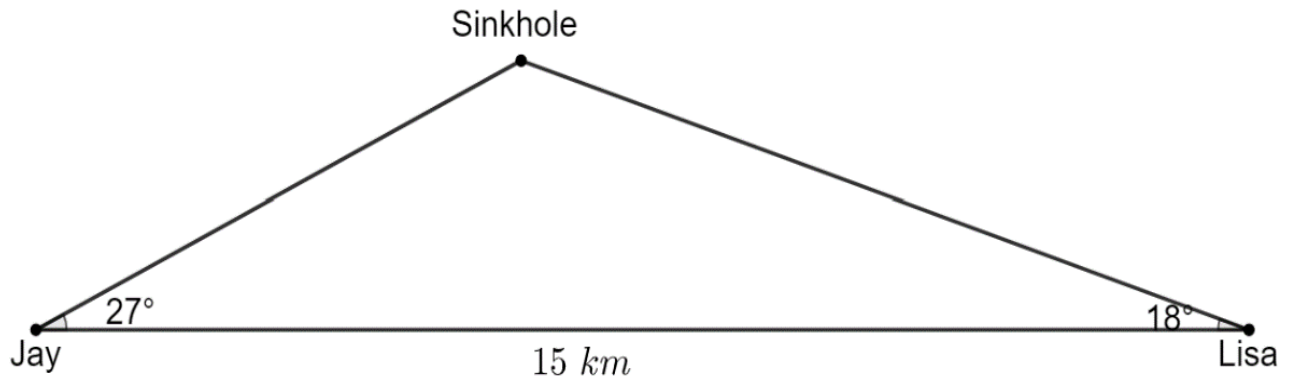
In $\triangle ZFS$, $m\angle SZF = 47^\circ$, $ZF = 28.3$ and $SF = 51.4$. Determine the measurement of each part of $\triangle ZFS$. Show your work below the table. If necessary, round to the nearest tenth.

	Part	Measurement
4.	$m\angle FSZ$	23.7°
5.	$m\angle SFZ$	109.3°
6.	$\overline{SZ}$	66.4°

$$\begin{aligned}
 4) \quad & \frac{\sin 47}{51.4} = \frac{\sin \angle FSZ}{28.3}; \quad \sin \angle FSZ = \frac{28.3 \sin 47}{51.4}; \quad \angle FSZ = 23.7^\circ \\
 5) \quad & m\angle SFZ = 180 - (47 + 23.475) = 109.255 \\
 6) \quad & \frac{\sin 47}{51.4} = \frac{\sin 109.255}{SZ}; \quad SZ = \frac{51.4 \cdot \sin 109.255}{\sin 47}; \quad SZ = 66.349
 \end{aligned}$$

Jay and Lisa are both hiking in the Leon Sinks Geological Area. They are currently 15 kilometers (km) apart. They plan to meet up at one of the sinkholes in the park.

7. How far, to the nearest hundredth of a kilometer, does Jay hike?



Sinkhole angle: $180 - (27 + 18) = 135^\circ$

Jay to Sinkhole: $\frac{\sin 18}{x} = \frac{\sin 135}{15}$; $x \approx 6.56$ km

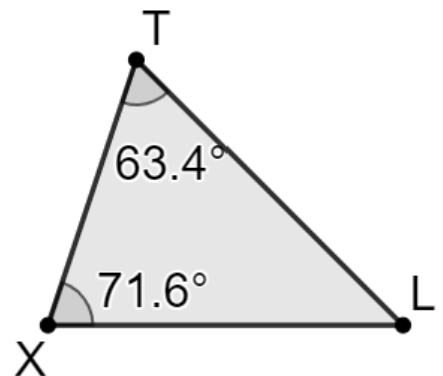
8. How far, to the nearest hundredth of a kilometer, does Lisa hike?

$\frac{\sin 27}{y} = \frac{\sin 135}{15}$; $y \approx 9.63$ km

Shanice runs in a competitive obstacle course. The obstacle course and some of its measurements are shown. The distance between point T and point X is 3.2 miles.

9. Write an equation that can be used to determine the distance, in kilometers, between point T and point L .

$\frac{\sin 71.6}{x} = \frac{\sin(180-63.4-71.6)}{5}$; (3.2 miles = 5 km)



10. Determine the perimeter, in miles, of the obstacle course.

TL : $\frac{\sin 71.6}{x} = \frac{\sin(180-63.4-71.6)}{3.2}$; $TL \approx 4.294$

XL : $\frac{\sin 45}{3.2} = \frac{\sin 63.4}{y}$; $XL \approx 4.046$

Perimeter: $3.2 + 4.294 + 4.046 = 11.54$ miles